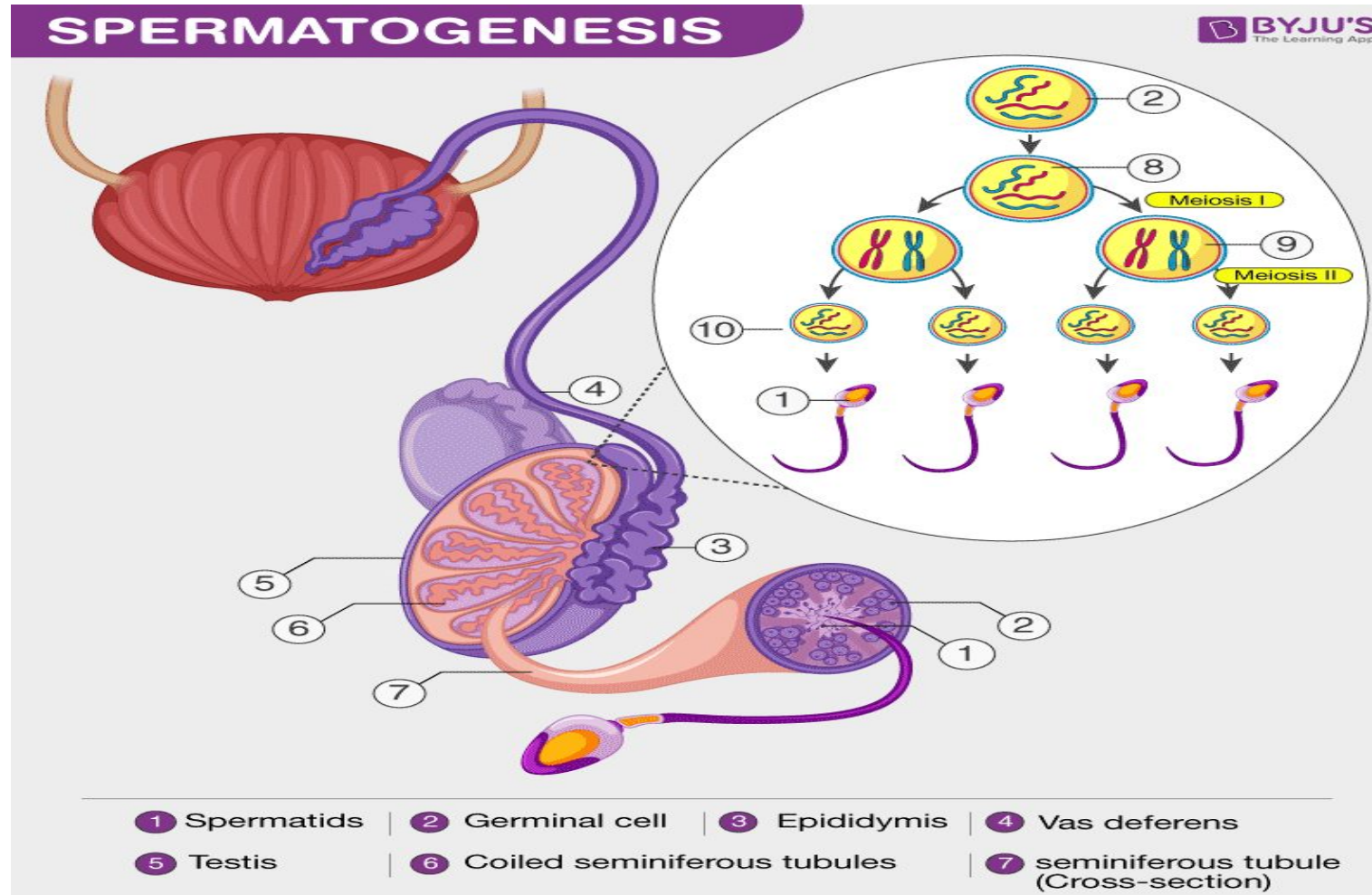


SPERMATOGENESIS



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LEARNING OBJECTIVES

At the end of lecture students should be able to recall the process of spermatogenesis:

- Describe the process of spermiogenesis
- Restate the different parts of sperm
- Enumerate the different parts of male reproductive system
- Explain the structure of testis
- Define the structure of seminiferous tubules
- Correlate the hormonal control with the development process.
- Illustrate the histologic features.
- Function of testosterone.
- Hypogonadism.
- Hypergonadism.

Male Reproductive System

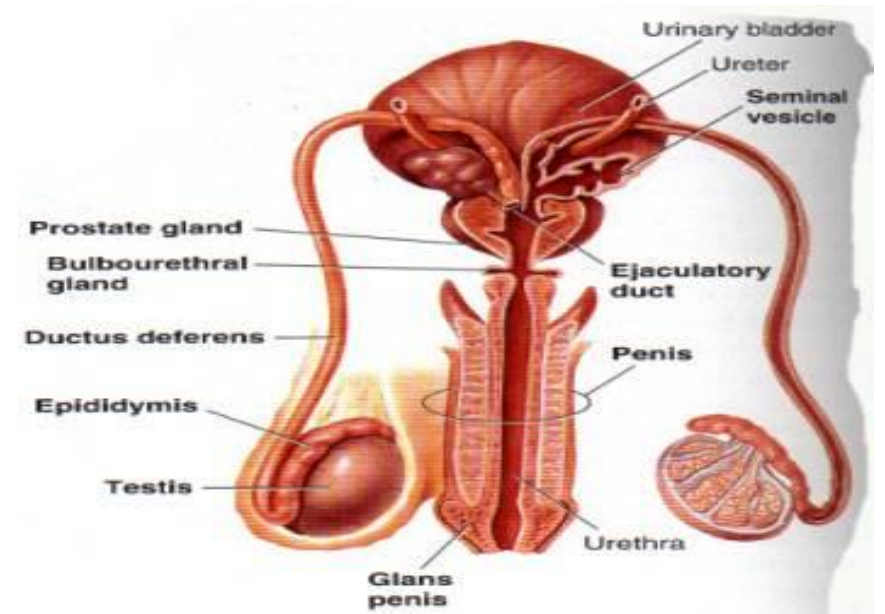
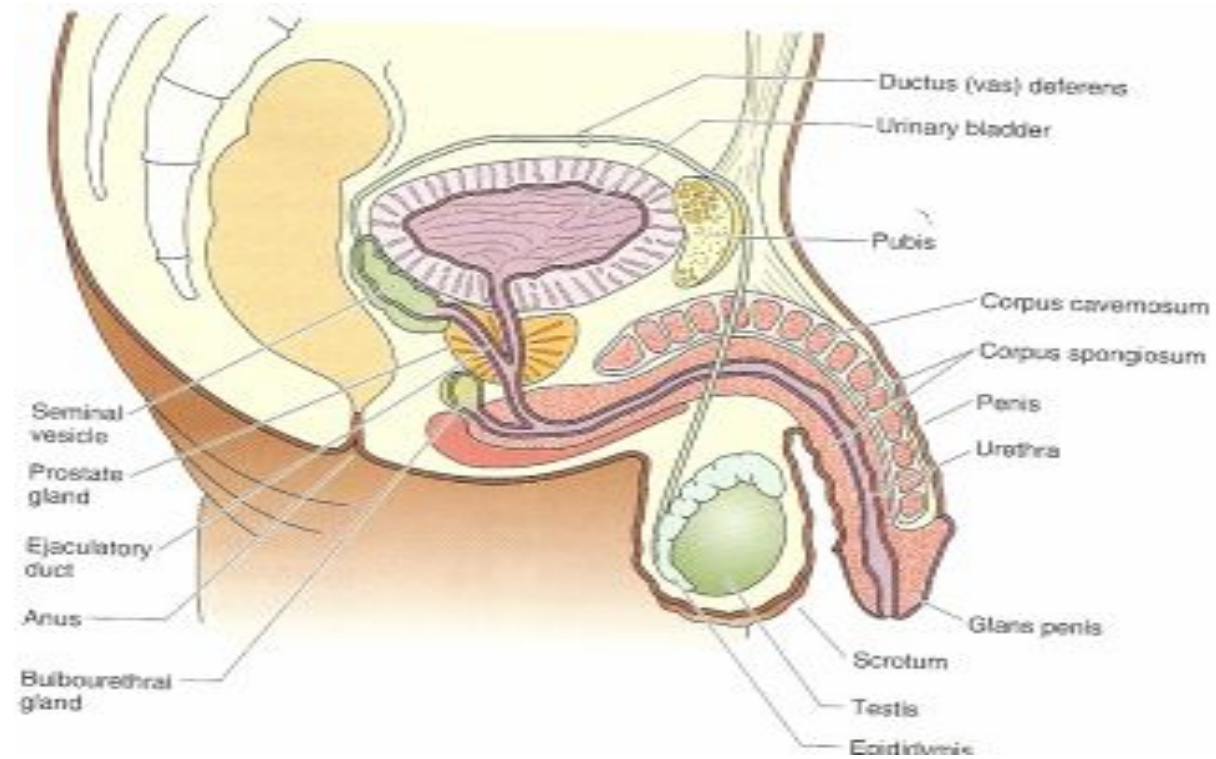
Functions of Reproductive system include:

- Production of gametes
- Storage gametes
- Nourishment gametes
- Transport gametes

MALE REPRODUCTIVE SYSTEM

Male reproductive system consists of:

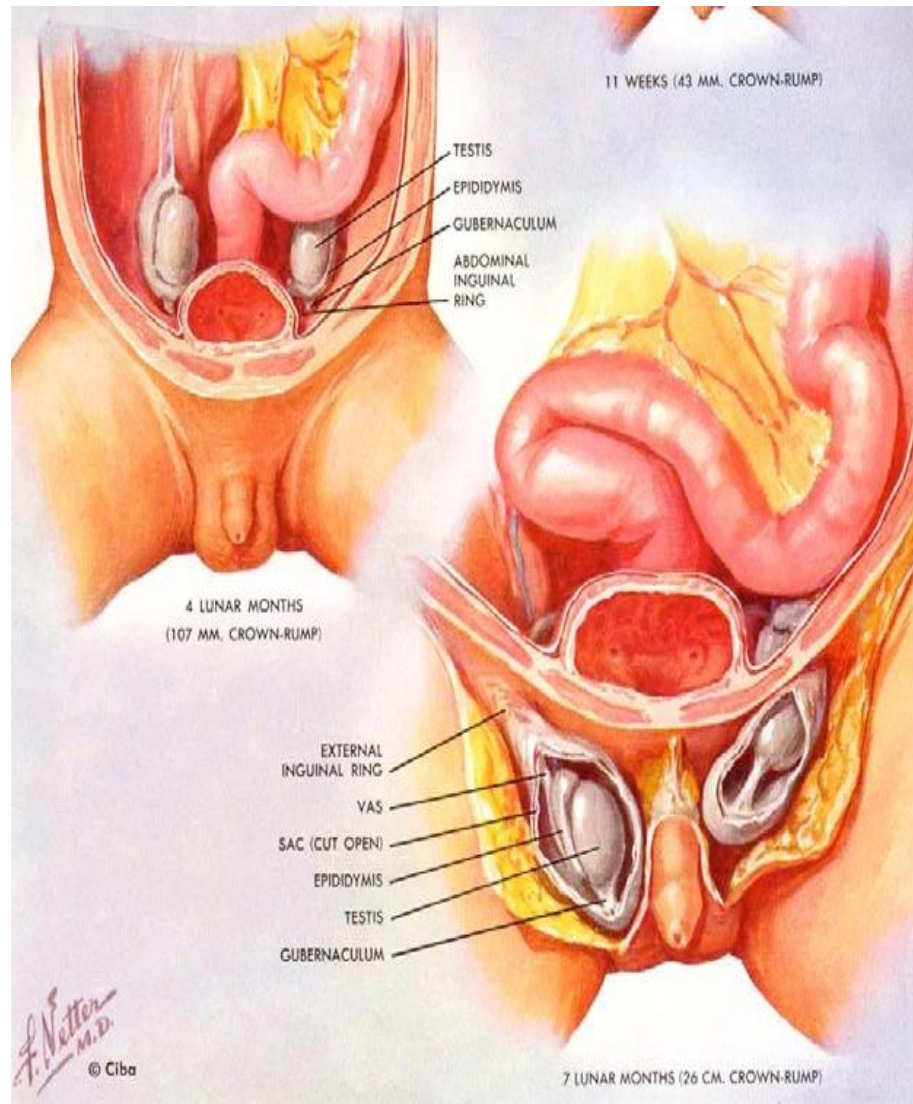
1. Testes suspended in scrotum
2. Genital ducts
 - a. Intra testicular
 - b. Extra testicular
3. Associated glands
 - a. Seminal vesicle
 - b. Prostate
 - c. Bulbourethral glands
4. Male copulatory organ, PENIS



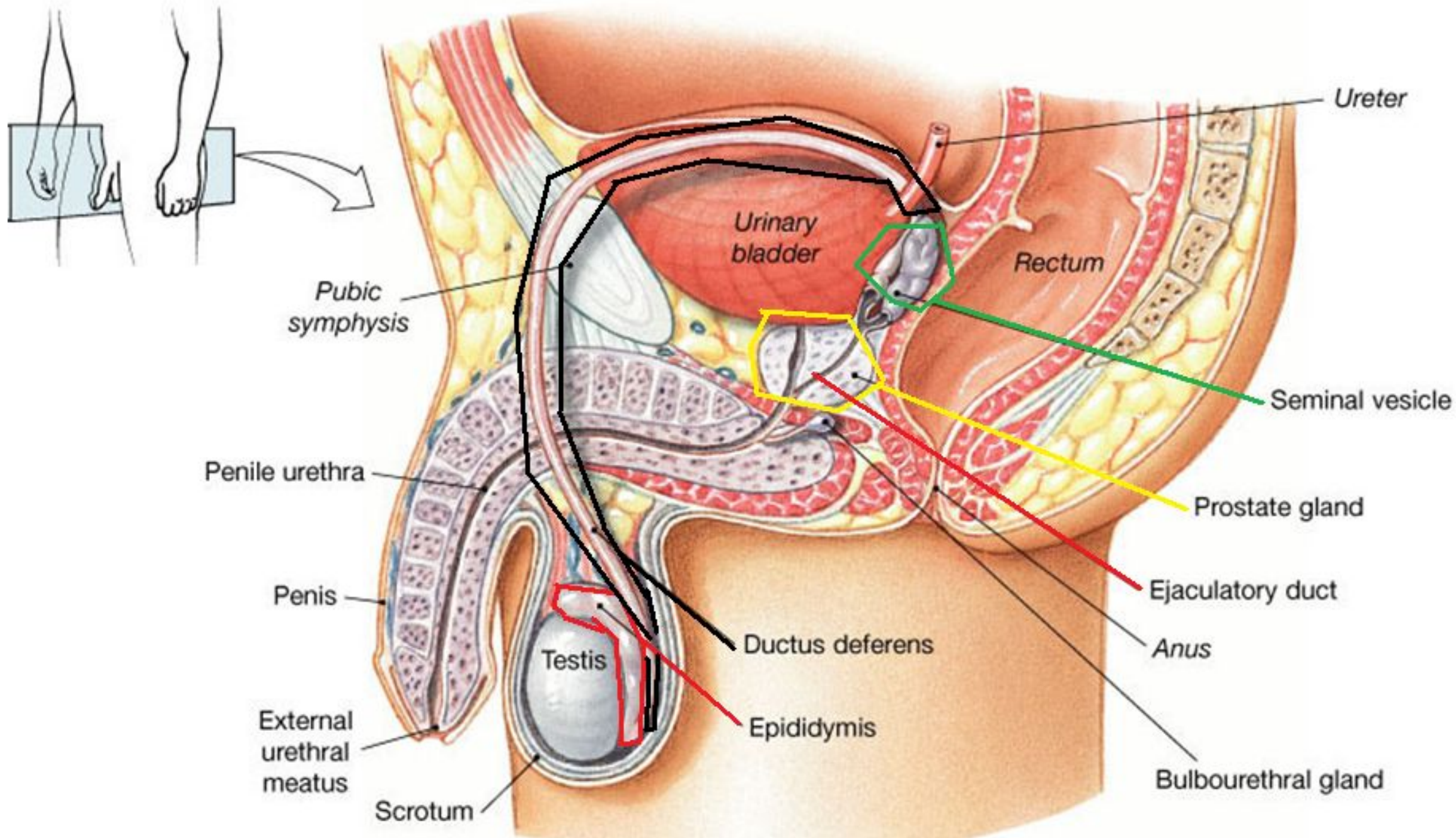
TESTES

- Oval shaped organ ,4 cm long x 2-3 cm wide and 3cm thick
- During embryogenesis testes develop retroperitoneal on the posterior wall of abdominal cavity, as they descend into scrotum, carry a portion of peritoneum out pocketing tunica vaginalis- forms serous cavity that partially surround the anterolateral aspect of each testes,permitting it some degree of mobility within its compartment in the scrotum.
- It has dual function that it act as exocrine gland by producing sperm
- Endocrine function is production an hormone the Testosterone

Testicular development



Abdominal cavity
from the medulla
of the primordial
gonad



GENERAL STRUCTURE

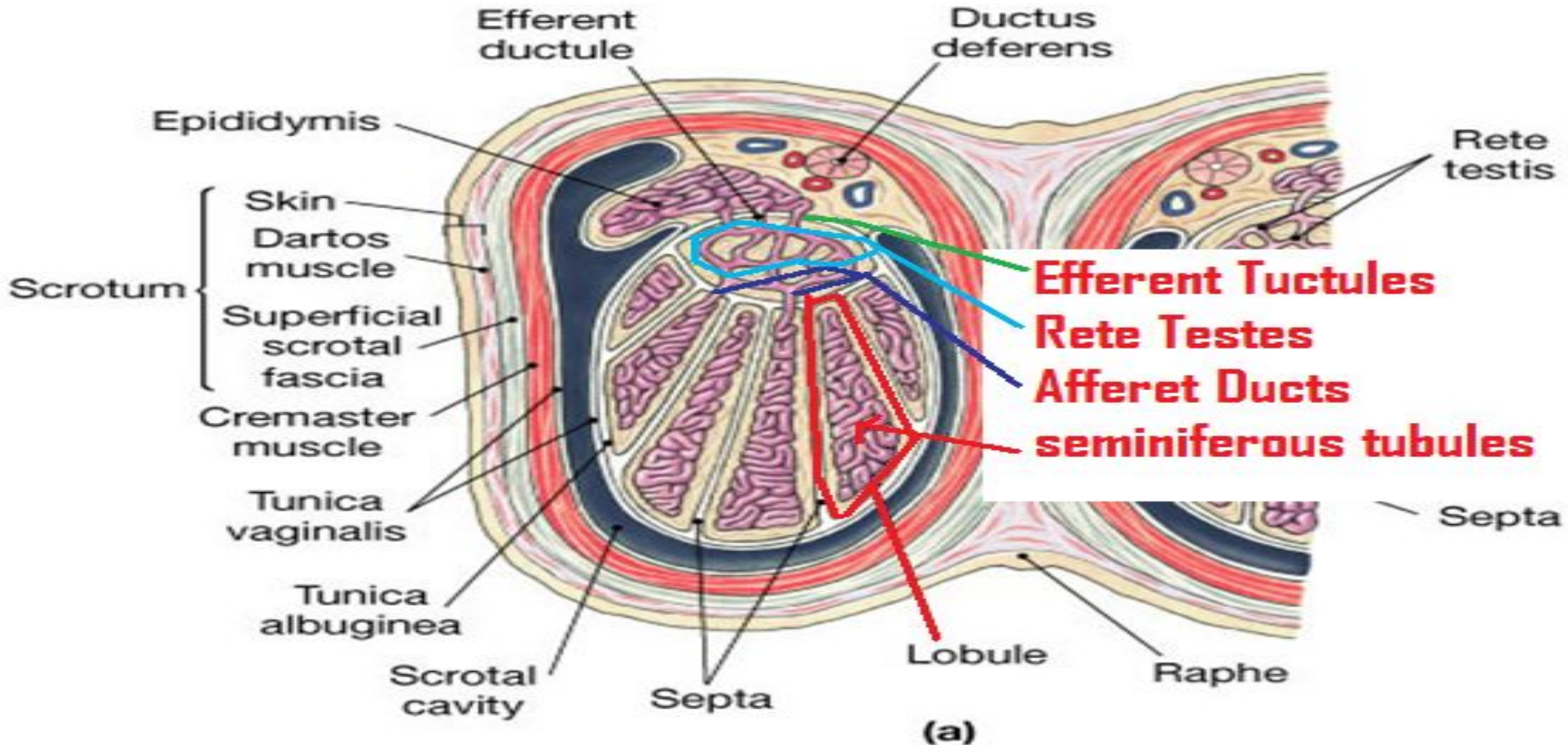
Testis □ constitutes three coats

□ **Tunica vaginalis**- the pouch of serous membrane that covers the testes.

□ **Tunica albuginea**- dense, irregular, collagenous connective tissue

□ **Tunica vasculosa**- forms vascular capsule of test situated beneath the Albuginea

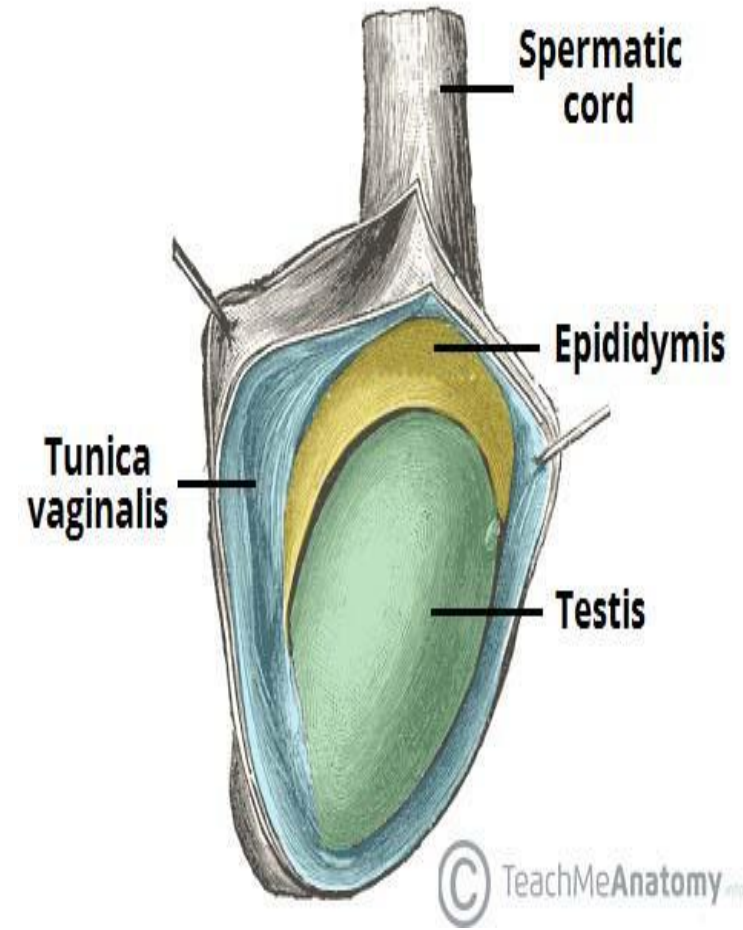
The Structure of the Testes



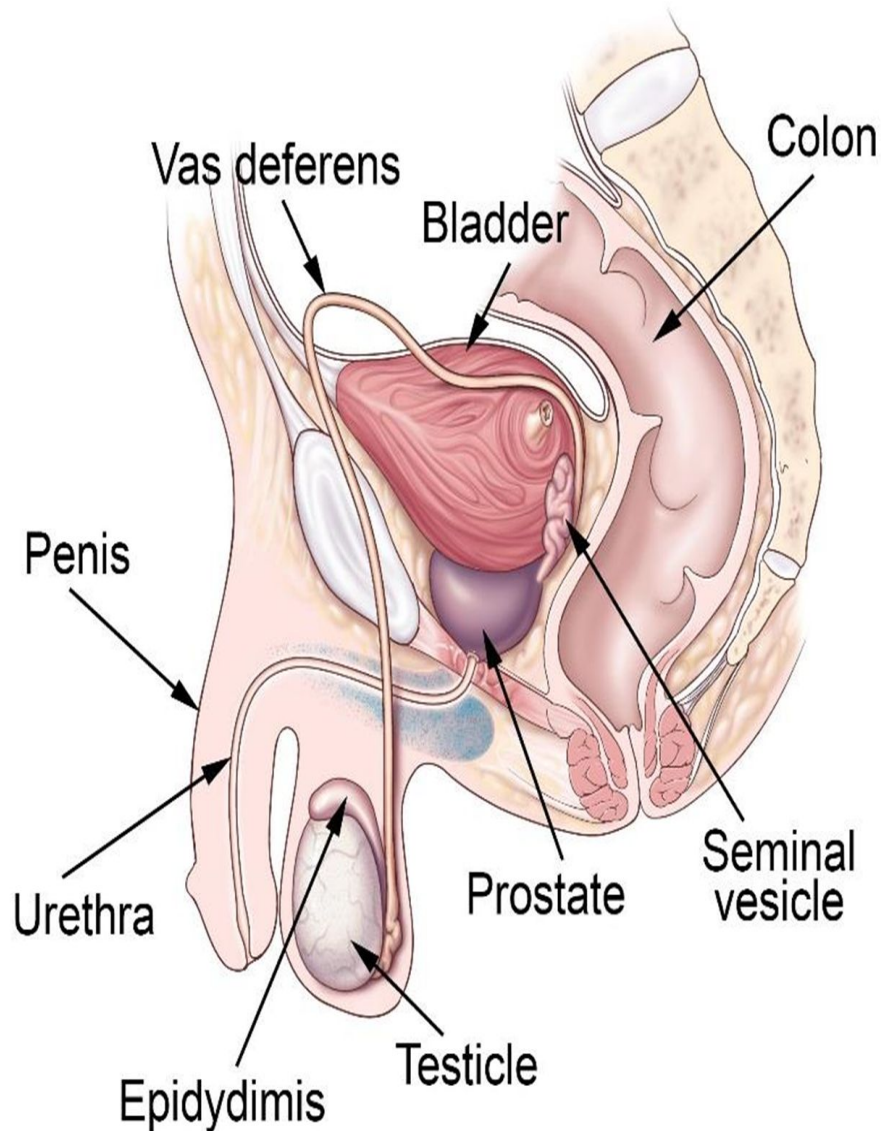
- The two testicles are egg-shaped male reproductive glands.
- They are suspended in a pouch of loose skin called the scrotum below and behind the penis.
- In maturity, they are usually slightly smaller than a golf ball. It is normal for one testicle to be larger than the other
- The testicles produce the hormone testosterone as well as sperm, the male reproductive cells.

Leydig cells: Secrete testosterone

Tunica albuginea: A dense capsule enclosing each testis; inhibits direct extension of tumor



Ducts in males



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- **Seminiferous tubules:** Small tightly coiled tubes inside the testes where sperm are produced
- **Rete testis:** branched network of ducts in the testis.
- **Epididymis:** Where sperm are matured and stored .
- **Ductus deferens:** also called vas deferens, that transports sperm cells from the epididymis, where the sperm are stored prior to ejaculation.
- **Ejaculatory duct:** A short connector from vas deferens to urethra.

Accessory glands and secretions

Seminal vesicles: Glands that secrete fructose and prostaglandins into the semen

Prostaglandins: Locally acting chemical messengers that stimulate contraction of the female reproductive system to assist in sperm movement.

Prostate Gland: Secretes a milky, alkaline fluid that neutralizes acidic vaginal secretions and maintains sperm.

Bulbourethral glands: Secretes a mucus like substance that provides lubrication for intercourse.

Semen: A mixture of sperm and various glandular secretions containing 5% spermatozoa.

Spermatogenesis

Introduction :

- The process of producing mature sperm is called spermatogenesis. Sperm are produced in the seminiferous tubules of the testes and become mature in the **epididymis**.
- The seminiferous tubules are lined by Sertoli cells. The entire process takes about 9 to 10 weeks.
- **Spermatogenesis** is the process by which haploid spermatozoa develop from germ cells in the seminiferous tubules of the testis.
- Production of sperm are Continuous and prolific.
- Each ejaculation contains ~100-650 million sperm

Sperm production: Spermatogenesis

Spermatogenesis involves three processes

1) Mitosis

In this process spermatocytes are produced from spermatogonium

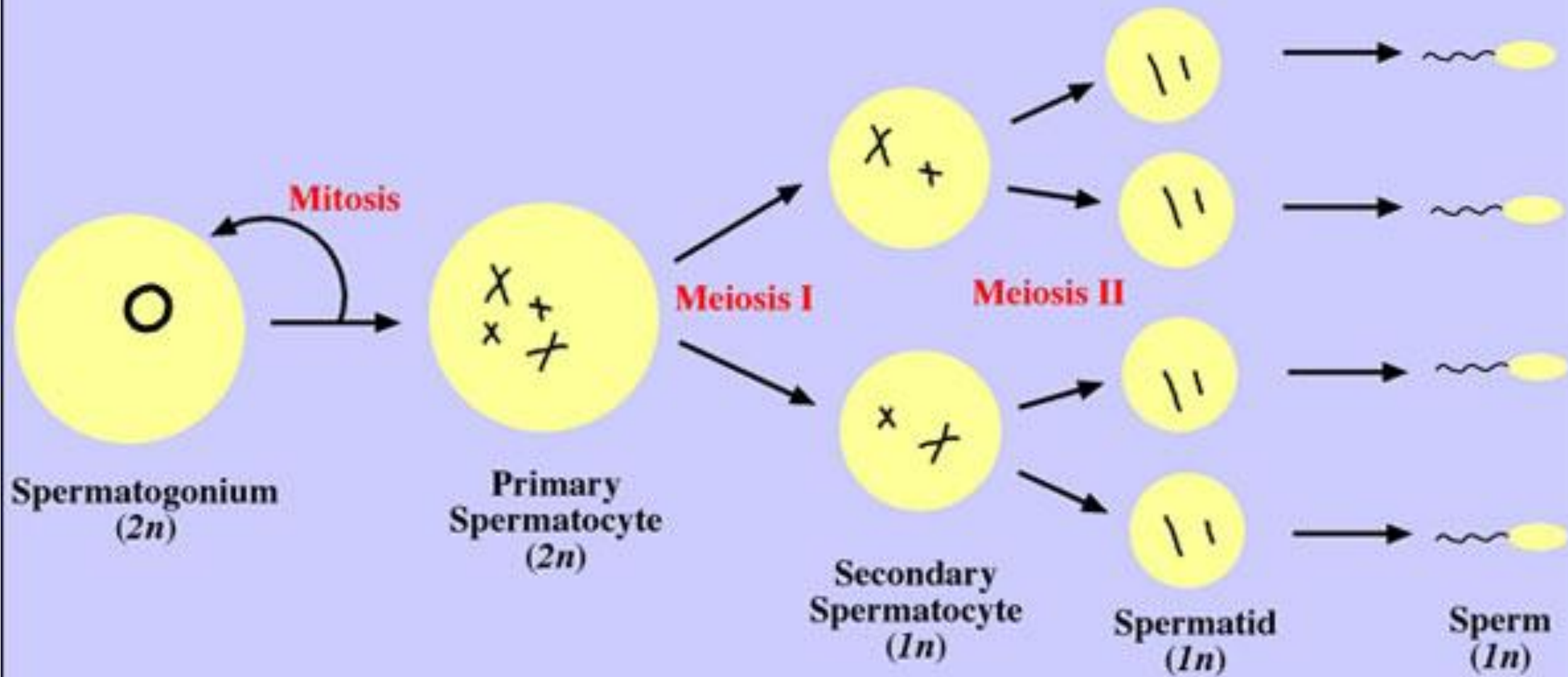
2) Meiosis

In this process spermatocytes go through meiosis I and meiosis II and produce 4 spermatids

3) Spermatogenesis

In this process spermatids differentiate into spermatozoa

Spermatogenesis



Spermatogenesis

Stages:

Mitosis

Meiosis

Spermio- genesis

Spermatogonia

One daughter cell remains at the outer edge of the seminiferous tubule to maintain the germ cell line

One daughter cell moves toward the lumen to produce spermatozoa

Primary spermatocyte

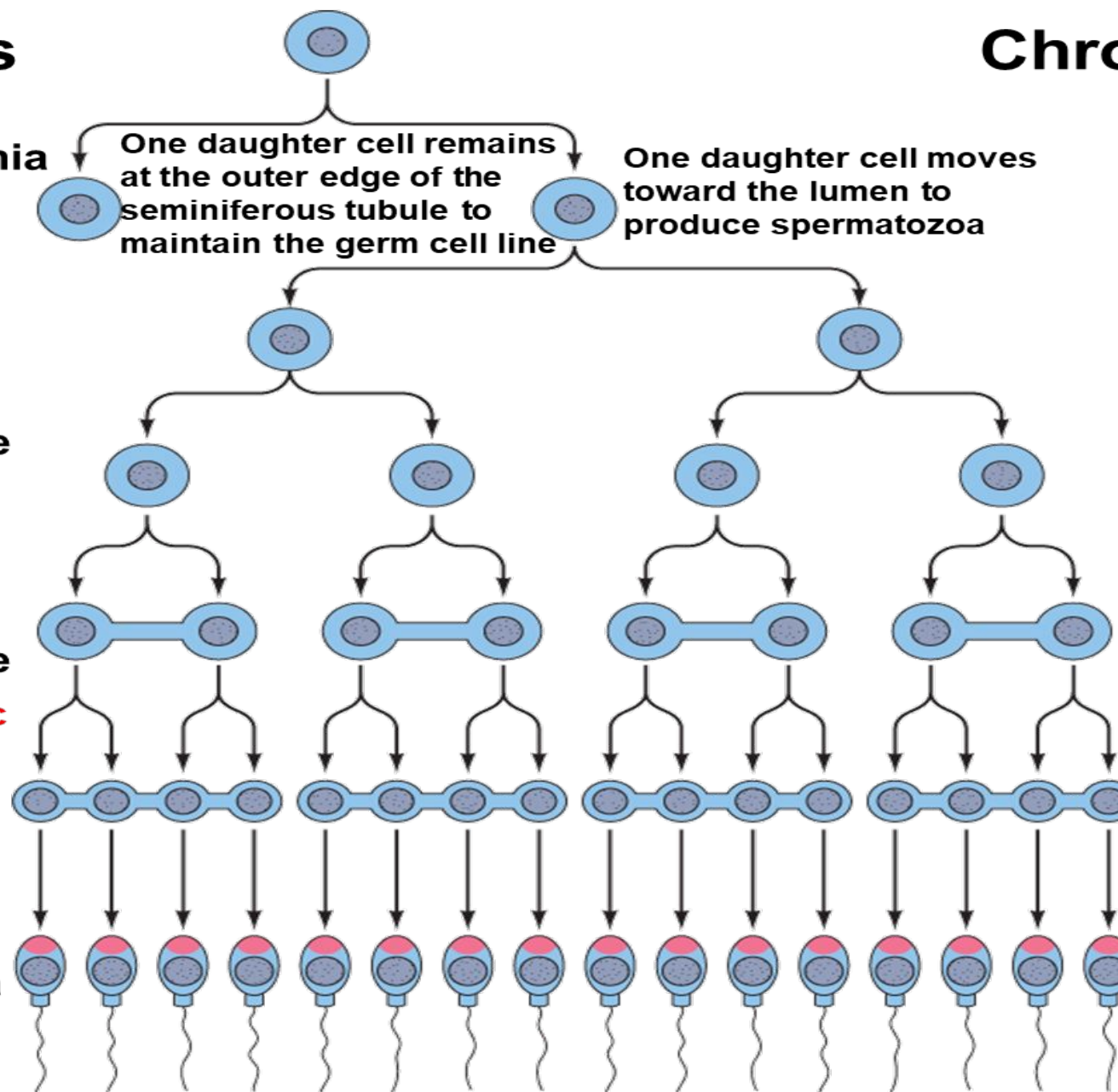
First meiotic division

Secondary spermatocyte

Second meiotic division

Spermatids

Spermatozoa



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Chromosomes:

46; 2n
(diploid number;
single strands)

46; 2n
(diploid number;
single strands)

46; 2n
(diploid number;
doubled strands)

23; n
(haploid number;
double strands)

23; n
(haploid number;
single strands)

23; n
(haploid number;
Single strands)

Male Reproductive Tract:

Epididymis:

- 1) Monitor and adjust fluid in seminiferous tubules
- 2) Store and protect spermatozoa
- 3) act as a recycling center for damaged spermatozoa
- 4) Facilitates functional maturation of spermatozoa

Duct Deferens:

Transport spermatozoa,
Store spermatozoa (for several months)

Ejaculatory duct:

Ejaculatory duct is a short passageway that penetrates the muscular wall of the prostate gland and empties into the urethra

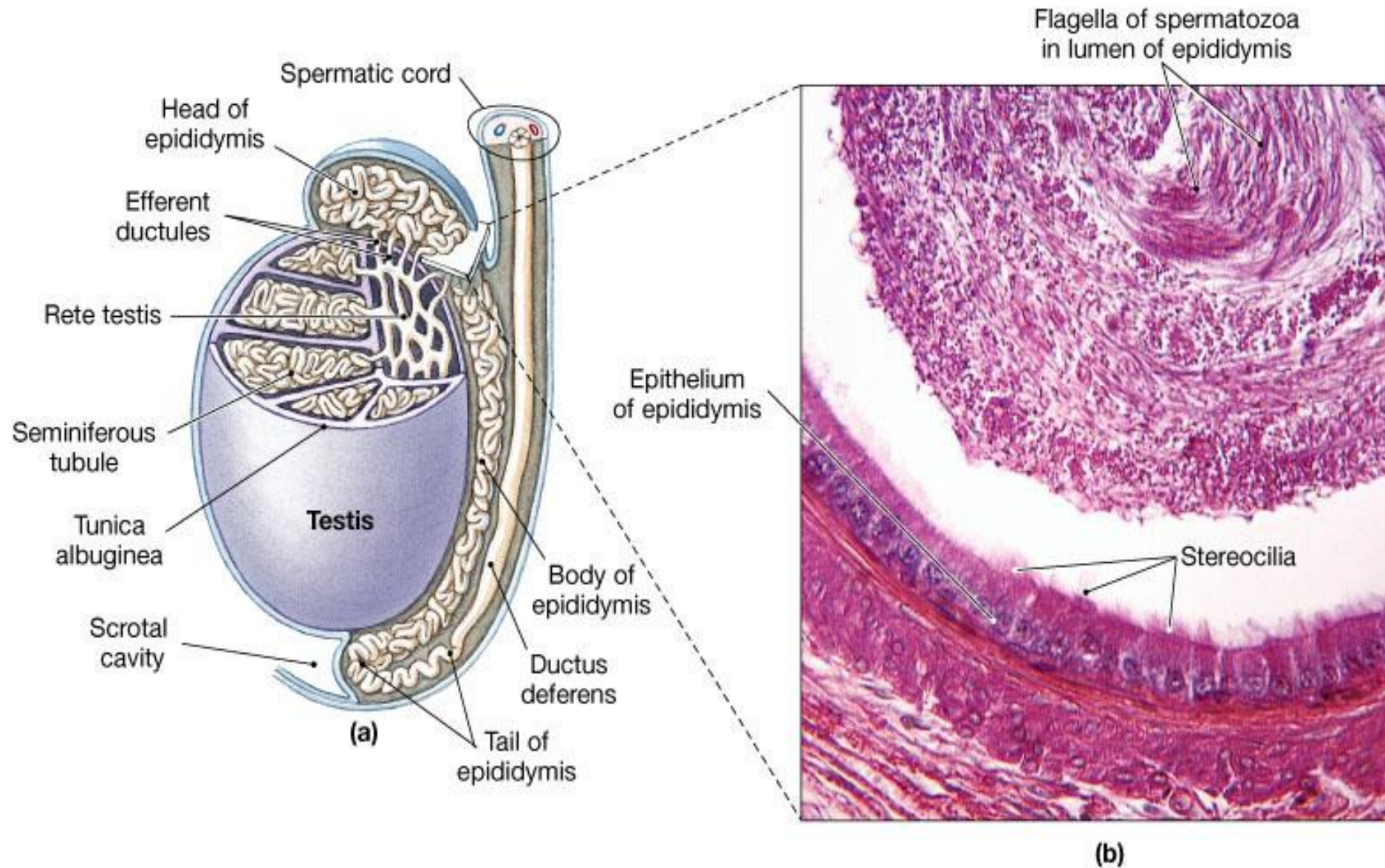
Urethra:

This passageway is used by both urinary and reproductive systems

Male Reproductive System: Sperm Passageway

1. Spermatozoa is produced in testes in the testicular part of seminiferous tubules
2. From seminiferous tubules sperms are passed to afferent ductless
3. Afferent ductules pass the sperm to rete testis
4. From rete testes sperm leave the testes by efferent tubules
5. Efferent ductules deliver the sperm to the epididymis
6. Epididymis pass the sperms to the duct deference
7. From ductus deferens, sperms are passed to the ejaculatory duct
8. Ejaculatory duct is connected to the urethra

The Epididymis



THE PATH OF SPERM

During ejaculation, sperm move from the male body as follows:

Vas deferens

Seminal vesicle

Ejaculatory duct

Prostate

Bulbourethral gland

Urethra

Epididymis

Bladder

Penis

Testes

Scrotum

1

MATURATION

Sperm mature in the epididymis.

2

STORAGE AND TRANSFER

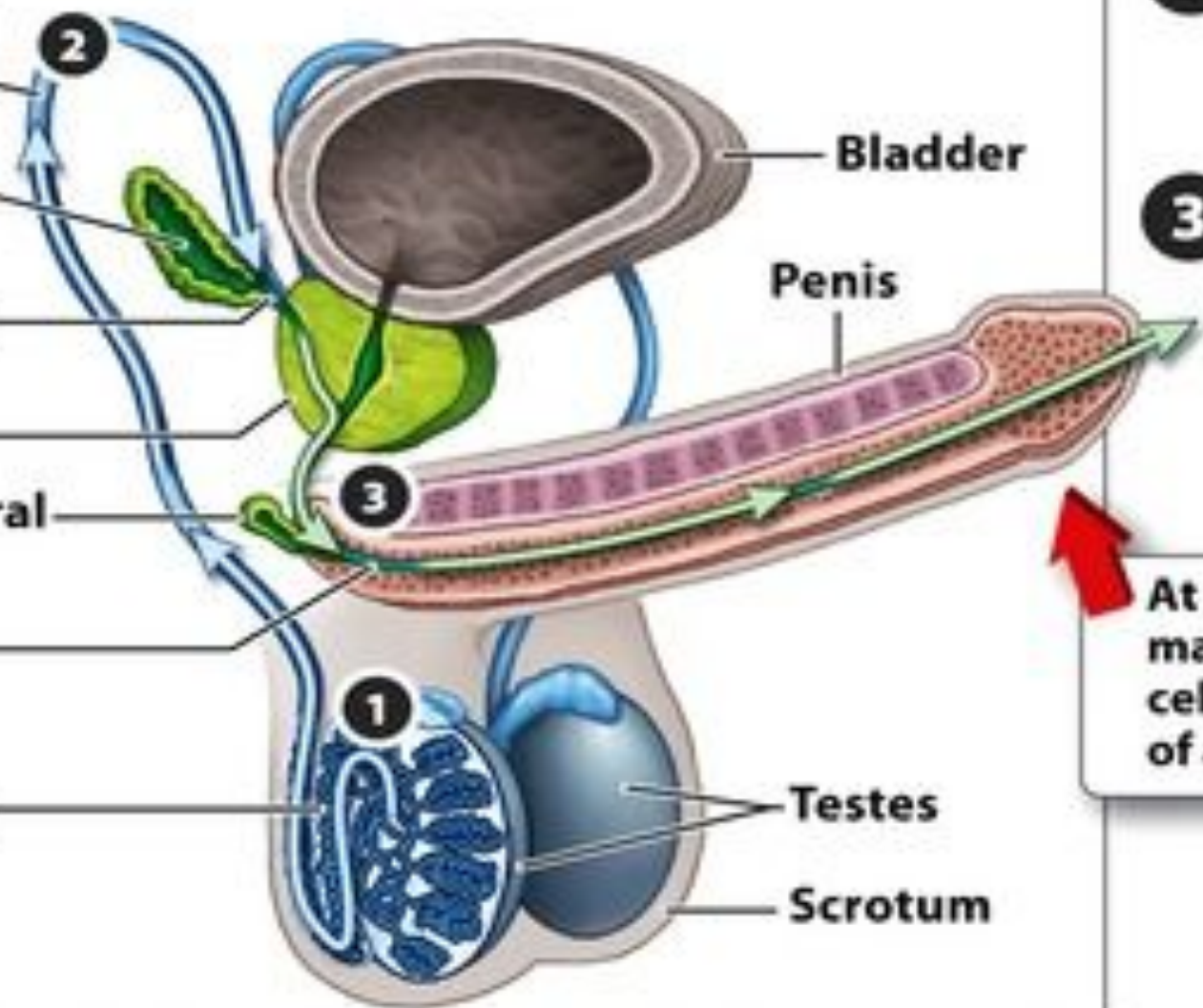
Muscle contractions cause sperm to move from the epididymis through the vas deferens.

3

DELIVERY

Sperm move through the ejaculatory duct into the urethra, where they can be expelled.

At ejaculation, approximately 300 million sperm cells are expelled as part of a fluid, called semen.



Male Reproductive System: Sperm Passageway

Rete testis

Vasa efferentia

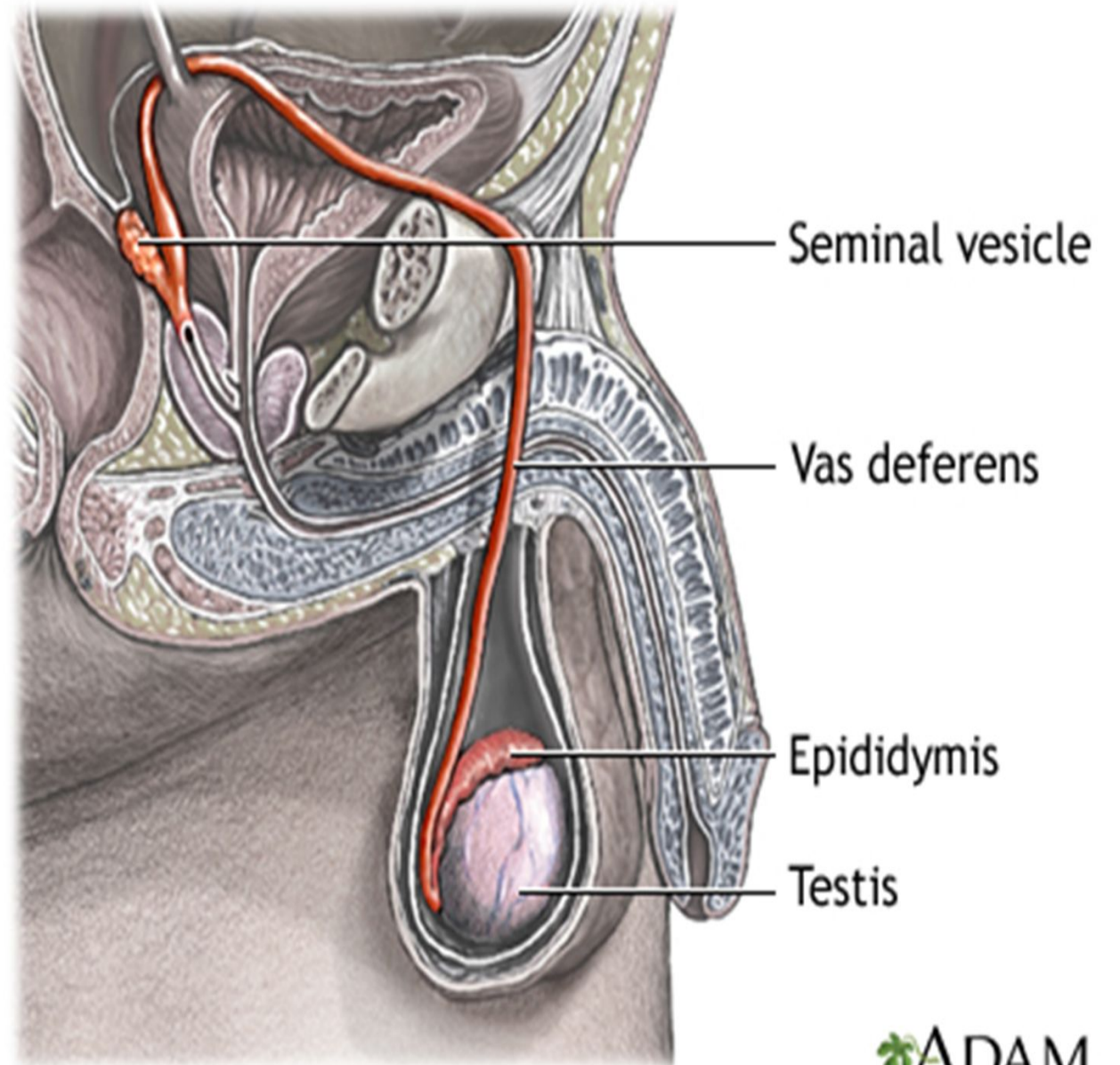
Epididymis

Vas deferens

Seminal Vesicles

Ejaculatory duct

Urethr



Anatomy of spermatozoon

Each spermatozoon is divided into 3 part:

1) Head that contains:

Nucleus and densely packed chromosomes

2) Middle piece:

that contains Mitochondria that produce the ATP needed to move the tail

3) Tail:

The only cell with flagellum in the human body

It enables the spermatozoa to swim

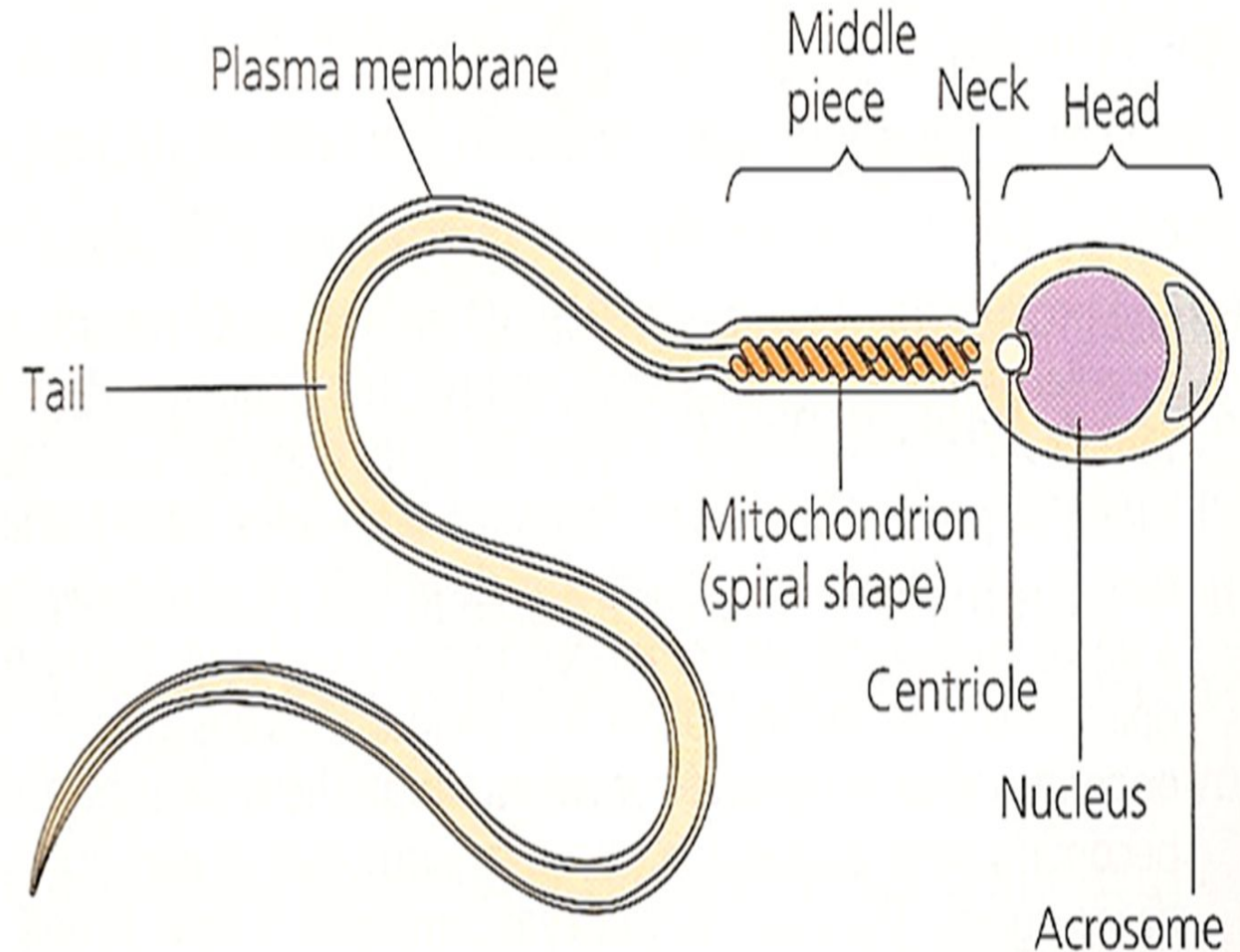


FIGURE 46.12 Structure of a human sperm cell.

The structure of sperm

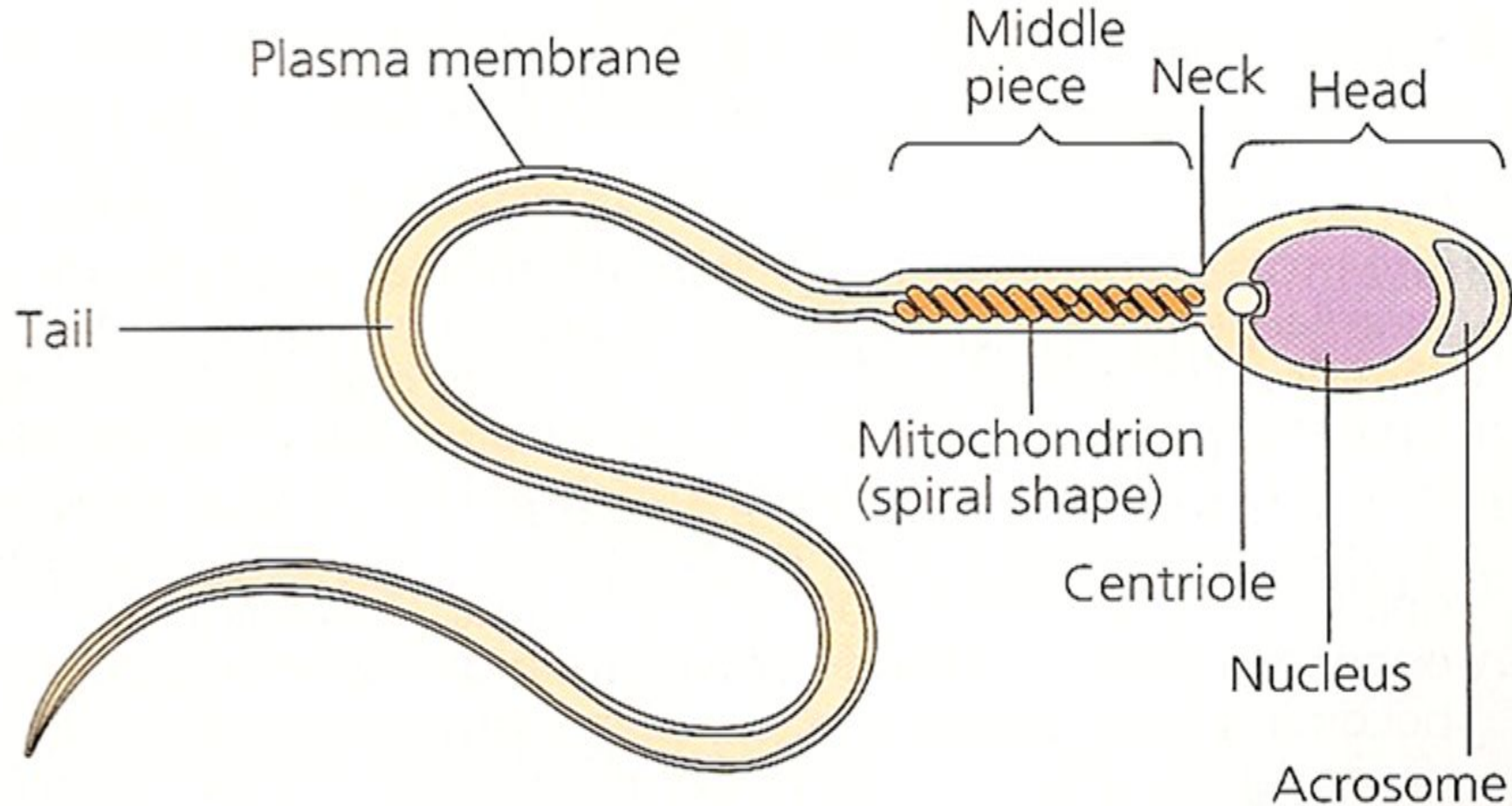
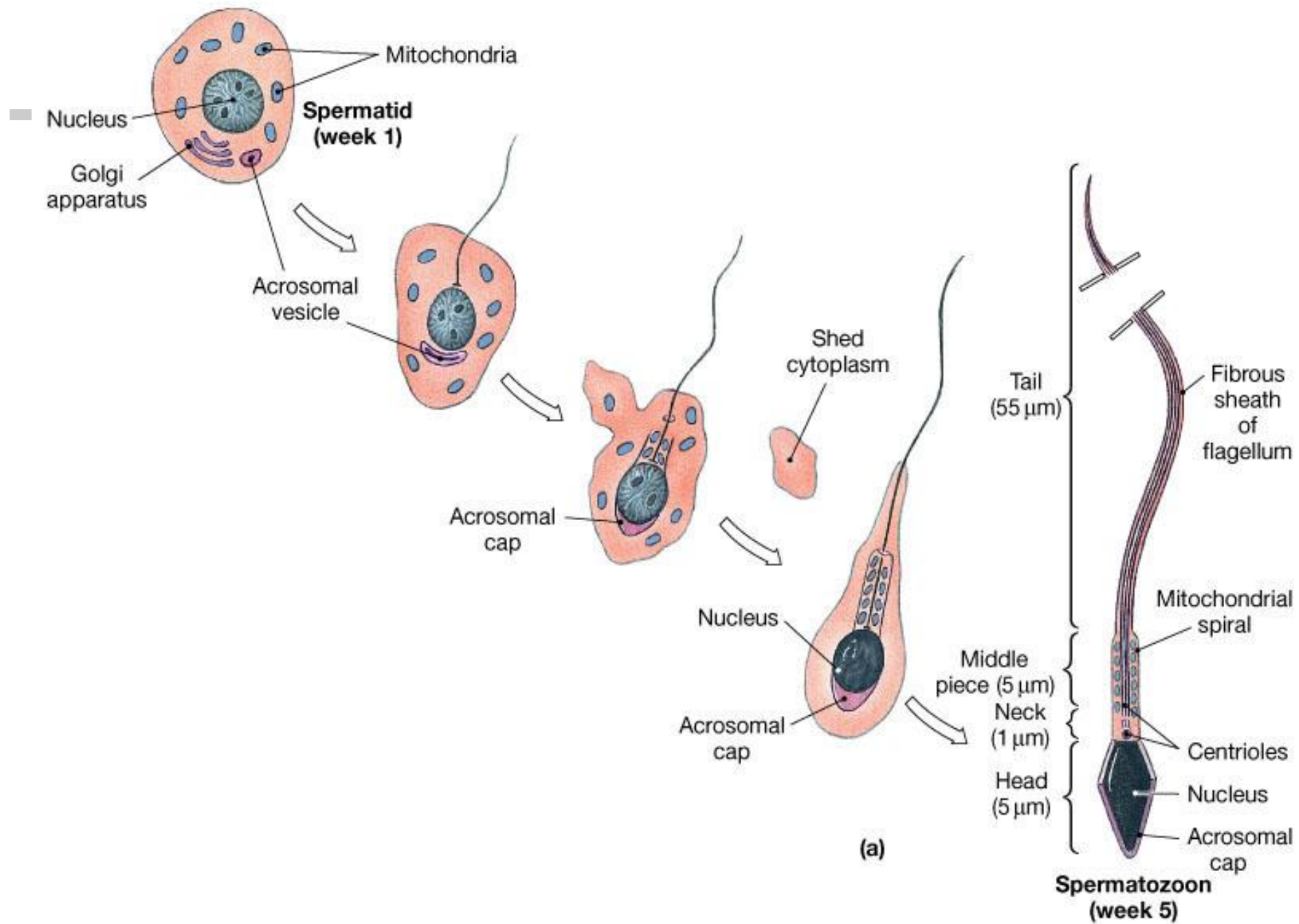


FIGURE 46.12 Structure of a human sperm cell.



Semen Contents

Semen is a mixture that is about 95% secretions and about 5% spermatozoa.

Typical ejaculate = 2-5 ml fluid that contain:

- 1) Contains between 20 – 100 million spermatozoa per ml
- 2) Seminal Fluid about 60%
- 3) Prostatic Fluid about 20-30 %
- 4) Enzymes such as:
 - Protease – helps dissolve mucous secretion in the vagina
 - Seminal plasmin – antibiotic enzyme that kills bacteria
 - Enzymes that convert fibrinogen to fibrin that will clot the semen
 - Fibrinolysis – liquefies the clotted semen after the vaginal environment is neutralized

Male Sex Hormones

The primary male sex hormone is testosterone, produced by the [testes](#), which is essential for developing male characteristics, sperm production, and maintaining muscle mass, bone density, and sex drive.

Other hormones, including follicle-stimulating hormone (FSH) and luteinizing hormone (LH) from the pituitary gland, are crucial for sperm production, and estrogen is also a male sex hormone, although it is produced from testosterone in small amounts.

Primary male sex hormone

Testosterone:

The main male sex hormone, it is produced in the Leydig cells of the testes.

Functions:

- 1-**Testosterone helps with sperm production, muscle and bone development, fat distribution, and sex drive.
- 2-**It also helps regulate red blood cell production and sexual function.
- 3-**Stimulate metabolism
- 4-**Stimulate the male secondary sexual characteristics.
- 5-**It is necessary for normal sperm production I,e stimulating spermatogenesis
- 6-**Testosterone is anabolic, meaning it builds up bone and muscle mass.
Anabolic effects include growth of muscle mass and strength, increased bone density and strength, and stimulation of linear growth and bone maturation.
- 7-**Androgenic effects include maturation of the sex organs, particularly the penis and the formation of the scrotum in the fetus, and after birth (usually at puberty) a deepening of the voice, growth of the beard and axillary hair.

Follicle stimulating hormone (FSH)

Produced by the pituitary gland, FSH is necessary for starting the process of sperm production.

Luteinizing hormone (LH)

Also produced by the pituitary gland, LH is needed to continue the process of sperm production.

Estrogen

While often associated with females, men also produce estrogen. In males, it is converted from testosterone and plays a role in the reproductive system, though in smaller amounts.

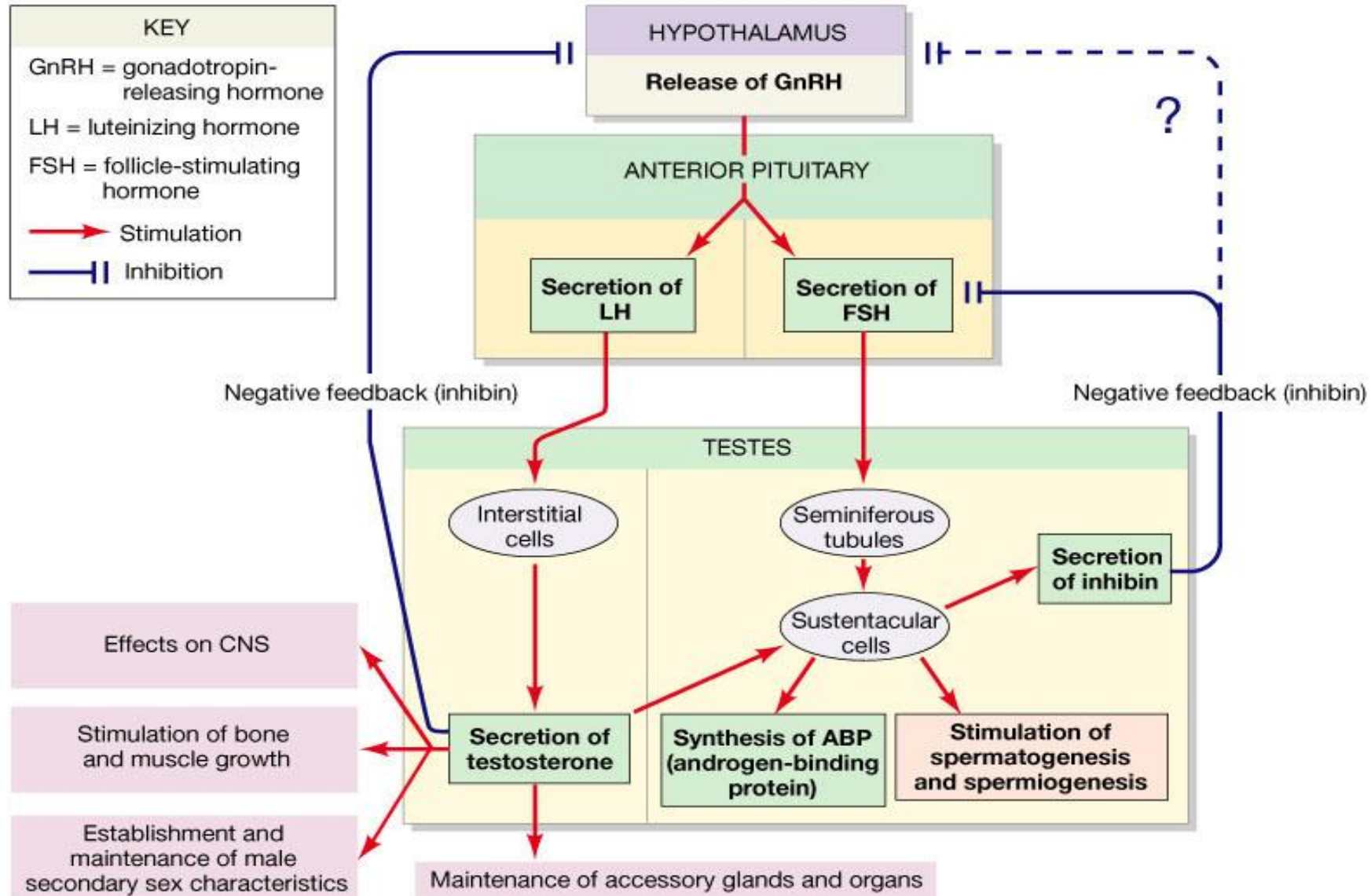
Dehydroepiandrosterone sulfate (DHEAS)

A male sex hormone that helps develop secondary male characteristics at puberty.

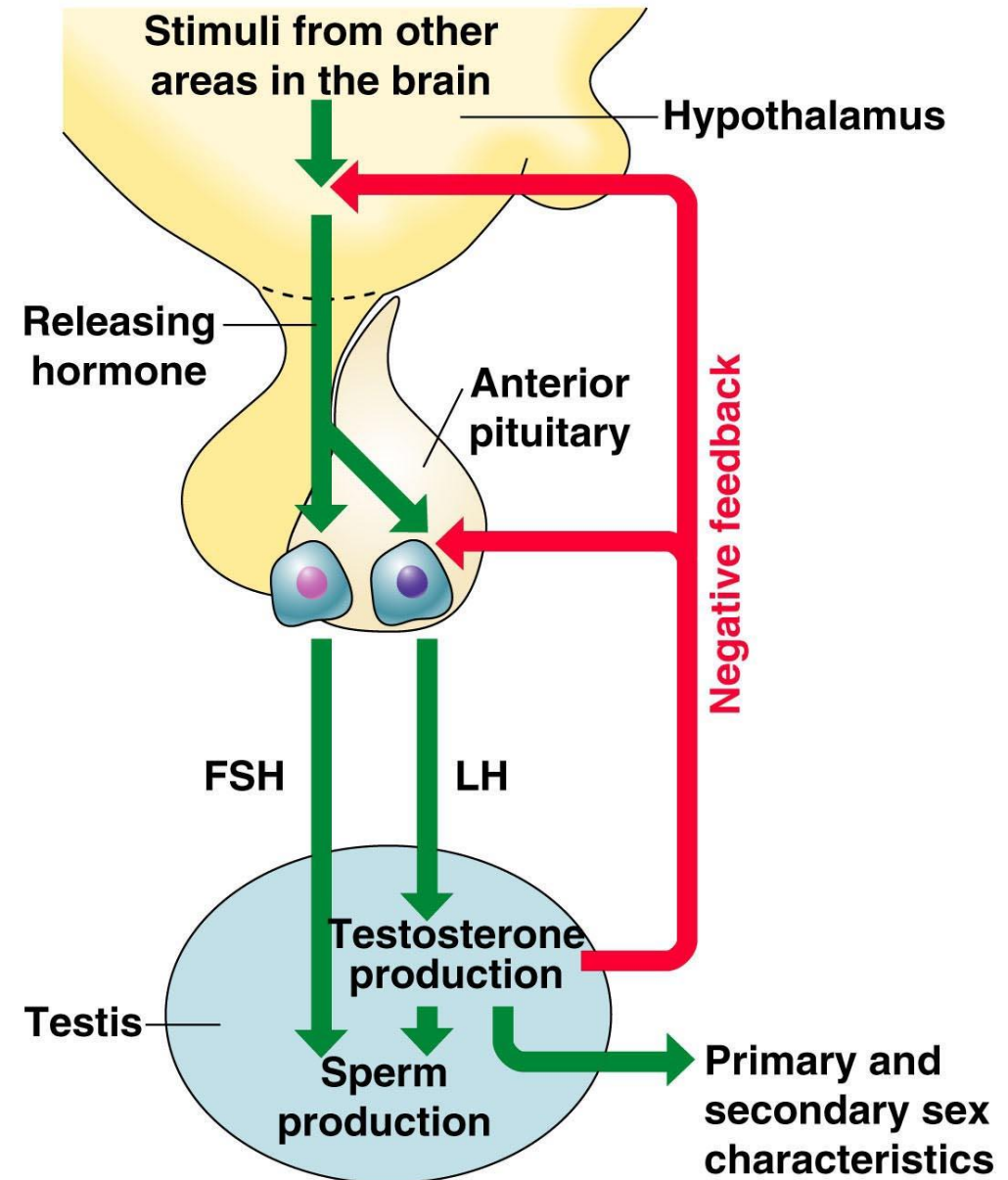
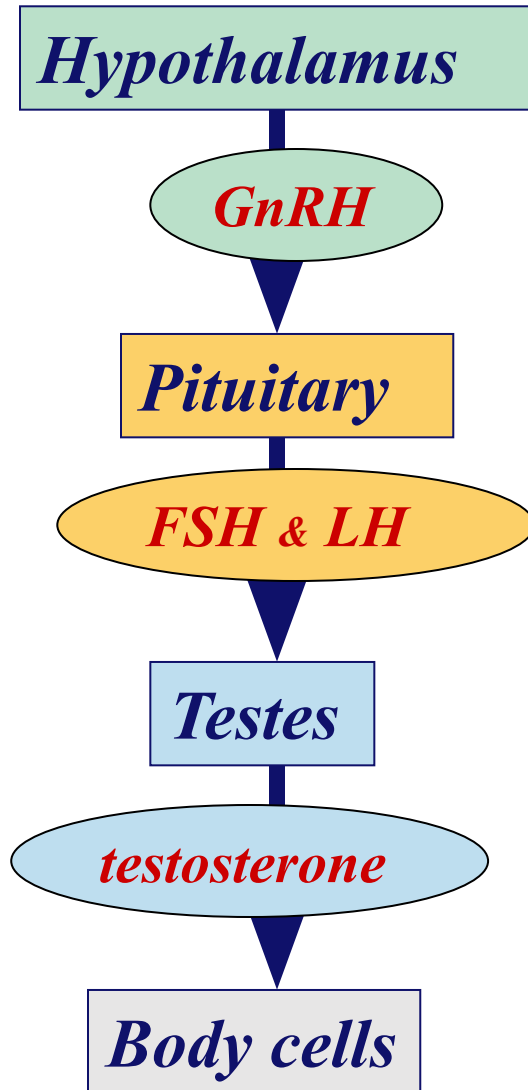
Hormonal control and function

- The production of male sex hormones is regulated by the hypothalamic-pituitary-gonadal (HPG) axis, a complex signaling pathway in the body.
- Hypothalamus: Releases gonadotropin-releasing hormone (GnRH), which signals the pituitary gland.
- Pituitary gland: In response to GnRH, the pituitary gland secretes luteinizing hormone (LH) and follicle-stimulating hormone (FSH).
- Testes: LH stimulates the Leydig cells in the testes to produce testosterone. FSH works with testosterone to stimulate the Sertoli cells, which promote sperm production (spermatogenesis).

Hormonal Feedback and the Regulation of the Male Reproductive Function



Sex hormone control in males



General roles of male sex hormones

Beyond reproduction, androgens are vital for overall health and development in all people.

- **Puberty:** Trigger the growth of sexual organs, development of body and facial hair, voice deepening, and increased muscle mass.
- **Sperm production:** Testosterone is essential for the production and maturation of sperm.
- **Bone health:** Androgens help maintain bone density and strength.
- **Red blood cell production:** They stimulate the production of red blood cells, which can contribute to a higher red blood cell count in males.
- **Libido and mood:** Androgens influence sex drive (libido) and contribute to overall mood and energy levels.

Hormonal Imbalances

An imbalance in male sex hormone levels can lead to a variety of health issues.

- **Low testosterone (Hypogonadism):**

Can cause low sex drive, fatigue, erectile dysfunction, decreased muscle mass, and increased body fat.

- **High androgens (Hyperandrogenism):**

While less common in males, high androgen levels can sometimes be a sign of a tumor. In females, it can cause conditions like Polycystic Ovary Syndrome (PCOS).